Data 37711 Homework 2 — Problem 1

Heteroscedastic regression: modelling $\mathbb{E}[Y \mid X]$ and $\sqrt{\mathrm{Var}(Y \mid X)}$

Yiwen Chen

Throughout, $f_\mu(x) \approx \mathbb{E}[Y \mid X = x]$ and $f_\sigma(x) \approx \sqrt{\mathrm{Var}(Y \mid X = x)}$. The code is in `problem1.py`.

(a) Link function

Write the network output as $g(x) = (g_1(x), g_2(x)) \in \mathbb{R}^2$. The two outputs have different constraints:

- $f_\mu(x)$ is a mean and can be any real number, so we use the **identity link**

  $$f_\mu(x) = g_1(x).$$

  (Trip durations are positive, so exp or softplus could be used here as well, but the conditional mean is far from 0 for nearly all trips, so the identity link is simplest.)

- $f_\sigma(x)$ is a standard deviation and must be strictly positive. We use the **softplus link** plus a small floor $\varepsilon > 0$:

  $$f_\sigma(x) = \mathrm{softplus}(g_2(x)) + \varepsilon = \log\bigl(1 + e^{g_2(x)}\bigr) + \varepsilon.$$

softplus : $\mathbb{R} \to (0, \infty)$ is smooth, monotone, and grows only linearly for large inputs, so the gradients do not blow up the way they can with $f_\sigma = \exp(g_2)$. The floor ε stops f_σ from going to 0, which would make the loss below go to $-\infty$ if the model fit a single point exactly. An equally valid option is the exponential link $f_\sigma(x) = \exp(g_2(x))$, which says the network predicts $\log \sigma$.

(b) Loss function

Under the model $Y_i \mid X_i \sim \mathcal{N}\bigl(f_\mu(X_i),\, f_\sigma(X_i)^2\bigr)$, independently over i, the conditional density of one observation is

$$p(y_i \mid x_i) = \frac{1}{\sqrt{2\pi}\, f_\sigma(x_i)} \exp\left(-\frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2}\right).$$

The log-likelihood of the sample is

$$\log \prod_{i=1}^{n} p(y_i \mid x_i) = \sum_{i=1}^{n} \left[-\tfrac{1}{2}\log(2\pi) - \log f_\sigma(x_i) - \frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2}\right].$$

Maximizing the likelihood is the same as minimizing the average **Gaussian negative log-likelihood**:

$$\boxed{\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \left[\log f_\sigma(x_i) + \frac{\bigl(y_i - f_\mu(x_i)\bigr)^2}{2 f_\sigma(x_i)^2}\right] + \tfrac{1}{2}\log(2\pi)} \tag{1}$$

where θ are the network weights and the constant does not affect the optimum.

Interpretation. The squared error term is weighted by $1/f_\sigma^2$, so points the model considers noisy count less when fitting the mean. The $\log f_\sigma$ term penalizes predicting a large σ everywhere. For a fixed x, minimizing the per-point loss over $s = f_\sigma(x)$ gives

$$\frac{\partial}{\partial s}\Big[\log s + \frac{r^2}{2s^2}\Big] = \frac{1}{s} - \frac{r^2}{s^3} = 0 \implies s^2 = r^2, \qquad r = y - f_\mu(x).$$

Averaged over the noise in $Y \mid X = x$, the population minimizer is $f_\sigma(x)^2 = \mathbb{E}[(Y - f_\mu(x))^2 \mid X = x]$. Together with $f_\mu(x) = \mathbb{E}[Y \mid X = x]$ (from setting $\partial/\partial f_\mu = 0$), this is exactly the conditional variance. So (1) estimates both target quantities. If $f_\sigma \equiv \sigma$ is a constant, (1) reduces to the usual MSE loss for f_μ (up to scale), so this loss generalizes least squares.

(c) Fitting an MLP on the taxi data

Data and split. `taxi_trips.parquet` has $n = 523{,}554$ trips, 7 standardized features and the target `trip_duration` (minutes). The target is right-skewed: median 8.6, mean 10.4, standard deviation 7.75, 99th percentile 37.1, max 228 minutes. I split the data at random into 70% training (366,487), 15% validation (78,533) and 15% test (78,534) using a fixed seed. The validation set is used only for early stopping and learning-rate scheduling. The test set is used once, for the final numbers.

Model. An MLP $g : \mathbb{R}^7 \to \mathbb{R}^2$ with hidden layers $(256, 256, 128)$ and SiLU activations, followed by the links from (a) with $\varepsilon = 10^{-3}$. For stable optimization I train on the standardized target $\tilde{y} = (y - \bar{y})/s_y$ (training mean and standard deviation) and map back with $f_\mu = \bar{y} + s_y \tilde{f}_\mu$ and $f_\sigma = s_y \tilde{f}_\sigma$. The bias of the σ output is initialized so that $\tilde{f}_\sigma \approx 1$ at the start of training. The loss is (1), optimized with Adam (learning rate 10^{-3}, batch size 1024). The learning rate is halved when the validation NLL stops improving for 2 epochs. Training stops early after 6 epochs without improvement, and the weights with the best validation NLL are restored.

Baselines. Each baseline is also a Gaussian model $(\mu(x), \sigma(x))$, so all models are scored with the same metrics:

1. **Constant**: $\mu = \bar{y}_{\text{train}}$, $\sigma = s_{y,\text{train}}$ (ignores x).
2. **Linear + constant σ**: OLS for the mean; σ is the standard deviation of the training residuals.
3. **Homoscedastic MLP**: the same architecture and training procedure for the mean, with one learned global σ. This baseline isolates the gain from modelling $\sigma(x)$.

Model	RMSE (min)	MAE (min)	Gaussian NLL (nats)	Coverage of central PI 50%	90%	95%
Constant	7.746	5.699	3.466	0.523	0.941	0.958
Linear + const. σ	7.551	5.592	3.441	0.536	0.935	0.956
Homoscedastic MLP	**3.637**	**2.303**	2.711	0.672	0.932	0.953
Heteroscedastic MLP (ours)	3.754	2.368	**2.648**	0.602	0.938	0.961

Table 1: Test-set results ($n = 78{,}534$). NLL is $-\frac{1}{n}\sum \log \mathcal{N}(y_i; f_\mu, f_\sigma^2)$ in minutes (lower is better). Coverage is the fraction of y_i inside $f_\mu \pm z_\alpha f_\sigma$; a calibrated model gets the nominal level.

Results (test set).

- **The MLPs beat both simple baselines by a wide margin.** RMSE drops from about 7.6 to about 3.7 minutes. The linear model barely improves on the constant, because trip duration depends mostly on the *distance* between pickup and dropoff (roughly $\|(\text{pickup}) - (\text{dropoff})\|$), and a linear function of the coordinates cannot represent that.

- **Modelling $\sigma(x)$ improves the likelihood.** The heteroscedastic MLP has the best test NLL (2.648 vs. 2.711 for the same network with one global σ). This is the model's training objective and the right score for a probabilistic regressor. Figure 2 (left) shows that the predicted σ tracks the realized error: when the test points are binned by $f_\sigma(x)$, the RMSE in each bin is close to the predicted σ. The right panel shows that the noise grows with the expected duration, from about 2 minutes for short trips to about 15–20 minutes for hour-long trips. A single global σ cannot capture this.

- **The heteroscedastic MLP has slightly worse RMSE.** This trade-off is expected: the loss (1) down-weights residuals at points with large f_σ, so the mean is fit less aggressively where the data are noisy. Its training curve (Figure 1) also shows more overfitting (train NLL 0.46 vs. val 0.57 in standardized units), so stronger regularization (weight decay or dropout) would probably help.

- **The Gaussian assumption is only approximate.** Trip durations are right-skewed and heavy-tailed. Even the heteroscedastic model over-covers at the 50% level (0.60) and is close to nominal at 90–95%. In other words, the residuals have a sharper peak and heavier tails than a Gaussian. A Gaussian on $\log Y$, or a Student-t likelihood, would fit better, but we keep the normal model as the problem specifies.

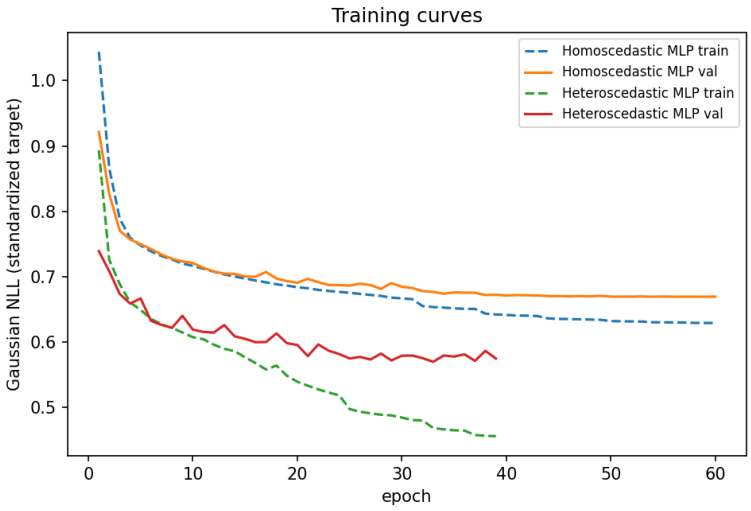


Figure 1: Training and validation NLL (standardized target) for the two MLPs.

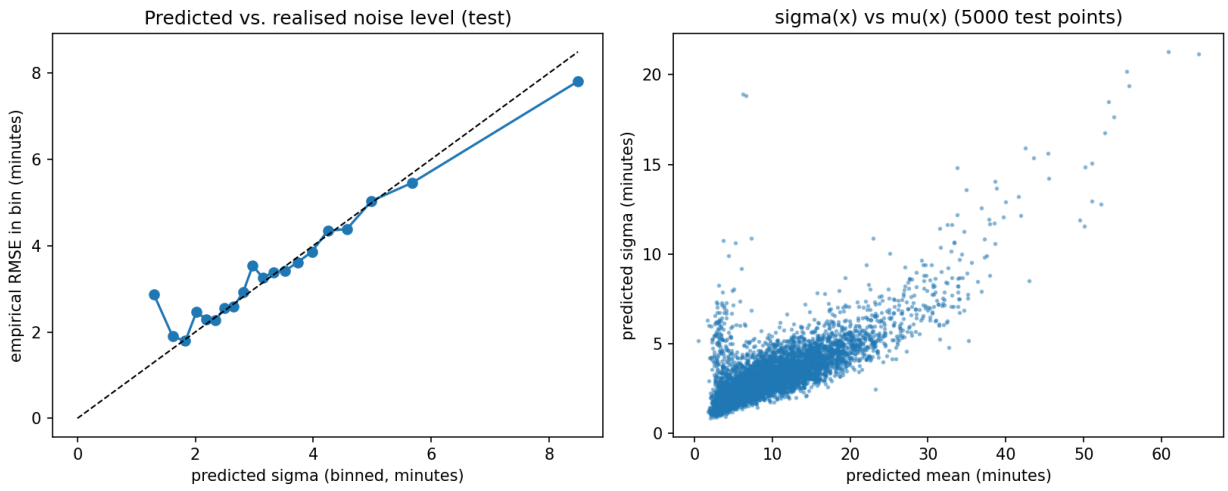


Figure 2: Left: test points binned by predicted $f_\sigma(x)$; empirical RMSE in each bin vs. mean predicted σ (dashed line = perfect). Right: $f_\sigma(x)$ against $f_\mu(x)$.

(d) Predicting $P(\textbf{trip} < 45\ \textbf{min})$

The fitted model gives a full predictive distribution $Y \mid X = x \sim \mathcal{N}\big(f_\mu(x), f_\sigma(x)^2\big)$. Therefore

$$\boxed{\hat{p}(x) = P(Y < 45 \mid X = x) = P\left(\frac{Y - f_\mu(x)}{f_\sigma(x)} < \frac{45 - f_\mu(x)}{f_\sigma(x)}\right) = \Phi\left(\frac{45 - f_\mu(x)}{f_\sigma(x)}\right)}$$

where Φ is the standard normal CDF. This uses both outputs. Two trips with the same predicted mean of 30 minutes get different probabilities: a trip with $f_\sigma = 5$ gets $\Phi(3) \approx 0.999$, and a trip with $f_\sigma = 15$ gets $\Phi(1) \approx 0.84$. A mean-only model cannot tell these two cases apart. To classify, predict "< 45" when $\hat{p}(x) \geq 0.5$. Because $\Phi(0) = 0.5$, this is equivalent to $f_\mu(x) \leq 45$: the 0.5-threshold decision depends only on the mean, while the probability itself depends on both f_μ and f_σ.

(e) Evaluation of the probabilities on the test set

For each test trip, the label is $z_i = \mathbf{1}\{y_i < 45\}$ and the prediction is $\hat{p}_i = \Phi\big((45 - f_\mu(x_i))/f_\sigma(x_i)\big)$. I report two proper scoring rules,

$$\text{Brier} = \frac{1}{n}\sum_i (\hat{p}_i - z_i)^2, \qquad \text{Log-loss} = -\frac{1}{n}\sum_i \big[z_i \log \hat{p}_i + (1 - z_i)\log(1 - \hat{p}_i)\big],$$

plus the accuracy of $\mathbf{1}\{\hat{p}_i \geq 0.5\}$ and the AUC. A natural extra baseline is the *majority-class predictor*, which always outputs the training frequency $\hat{p} = \hat{P}(Y < 45)$.

Model	Brier ↓	Log-loss ↓	Accuracy@0.5 ↑	AUC ↑
Majority class (train freq. 0.9956)	0.00444	0.02857	0.99554	0.500
Constant Gaussian	0.00446	0.05525	0.99554	0.500
Linear + const. σ	0.00446	0.04662	0.99554	0.843
Homoscedastic MLP	**0.00288**	0.03170	**0.99665**	0.952
Heteroscedastic MLP (ours)	0.00291	**0.02063**	0.99660	**0.959**

Table 2: Test-set evaluation of $\hat{p}(x) = P(Y < 45 \mid x)$. 99.55% of test trips (all but 350 of 78,534) are shorter than 45 minutes.

Discussion.

- **The classes are very imbalanced.** Only 350 of the 78,534 test trips (0.45%) take 45 minutes or longer. A predictor that always says "< 45" already reaches 99.55% accuracy, so accuracy on its own says little. The heteroscedastic MLP reaches 99.66% accuracy, which removes about 25% of the majority-class errors. At the 0.5 threshold it flags 161 trips as long, and 122 of them really are long. The constant and linear baselines never predict a long trip.

- **Proper scoring rules.** Both MLPs reduce the Brier score by about 35% relative to the baselines (0.0029 vs. 0.0044). On log-loss the heteroscedastic model is clearly best (0.0206). It is the only model that beats the trivial majority-class predictor (0.0286), and it improves on the homoscedastic MLP by 35%. The homoscedastic MLP is *worse* than the trivial predictor on log-loss even though its Brier score and accuracy are good. The reason is that with one global $\sigma \approx 3.6$ minutes, a trip predicted at $\mu = 25$ gets $P(Y \geq 45) = 1 - \Phi(5.5) \approx 2 \times 10^{-8}$. When such a trip does take more than 45 minutes, the log-loss charges about $\log(1/(2 \times 10^{-8})) \approx 18$ nats. The heteroscedastic model assigns larger σ to long and uncertain trips, so it spreads probability into the tail much more realistically. This is exactly where modelling $\sigma(x)$ matters.

- **Calibration.** Figure 3 shows that all Gaussian models are *over-confident* at small predicted probabilities. Trips assigned $P(Y \geq 45) < 10^{-3}$ actually take ≥ 45 minutes about 10^{-3}–10^{-2} of the time, because the right tail of trip durations is heavier than a Gaussian tail. The heteroscedastic model is the closest to the diagonal, and above about 0.05 it is well calibrated. A heavier-tailed or log-scale likelihood would reduce the remaining miscalibration.

- As shown in (d), the 0.5-threshold decision is $f_\mu(x) \leq 45$, so it depends only on the mean. That is why the two MLPs have nearly identical accuracy. The difference between them shows up in the probabilities, which is what the Brier score, log-loss and calibration measure.

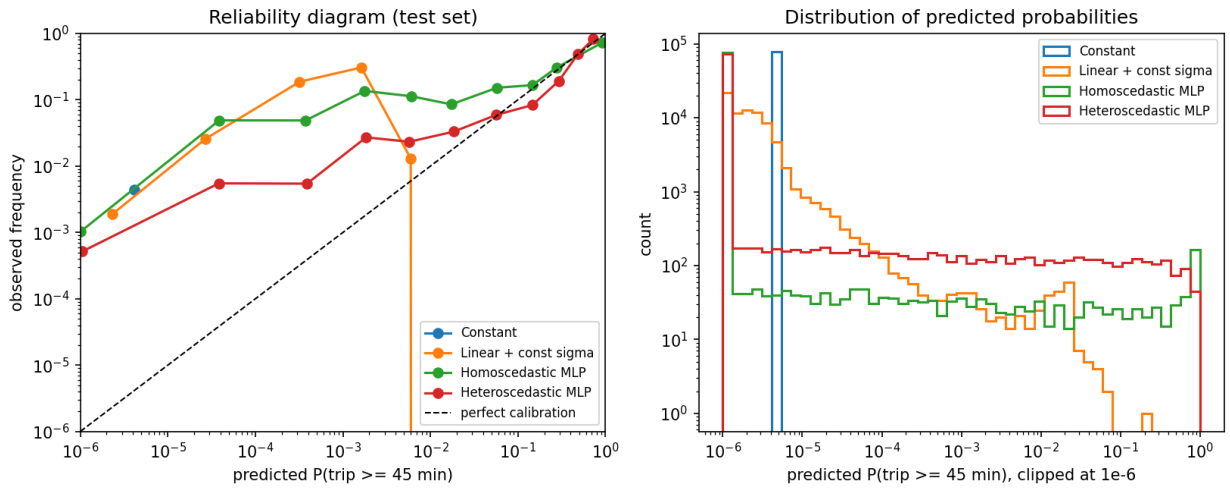


Figure 3: Left: reliability diagram for the rare event $\{Y \geq 45\}$ on the test set (log–log axes; dashed line = perfect calibration). Right: histogram of predicted $P(Y \geq 45)$, clipped at 10^{-6}.